

Database Systems

CSE 215

Lecture 03: SQL

2022

Single Table Query

SQL

- Data Definition Language (DDL)
 - Create/alter/delete tables and their attributes
 - Previous lectures...
- Data Manipulation Language (DML)
 - Query one or more tables – discussed next !
 - Insert/delete/modify tuples in tables

Table name

Attribute names

Tables in SQL

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

Tuples or rows

SQL Query

Basic form: (plus many many more bells and whistles)

<pre>SELECT <attributes> FROM <one or more relations></pre>

Simplest SQL Query

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

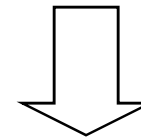
Show all entries from product table

Simplest SQL Query

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

```
SELECT *  
FROM Product
```



PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

“selection”

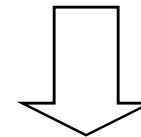
Single Column Selection

Show all product categories

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

```
SELECT category
FROM Product
```



“selection” and
“projection”

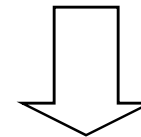
Category
Gadgets
Gadgets
Photography
Household

Single Column Selection (Eliminating Duplicates)

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

```
SELECT DISTINCT category  
FROM Product
```

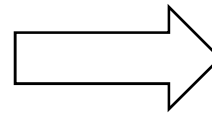


Show all product categories

Category
Gadgets
Photography
Household

In summary..

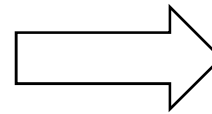
```
SELECT DISTINCT category  
FROM Product
```



Category
Gadgets
Photography
Household

Compare to:

```
SELECT category  
FROM Product
```



Category
Gadgets
Gadgets
Photography
Household

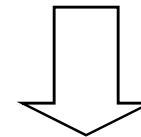
Multi-column Selection

List all products along with their price and manufacturers

Product

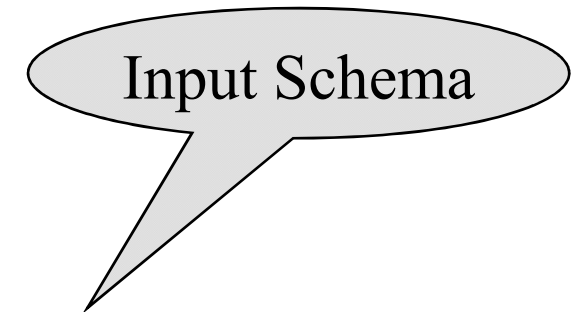
PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

```
SELECT pname, price, manufacturer
FROM Product
```

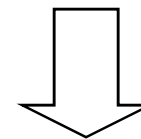


PName	Price	Manufacturer
Gizmo	\$19.99	GizmoWorks
Powergizmo	\$29.99	GizmoWorks
SingleTouch	\$149.99	Canon
MultiTouch	\$203.99	Hitachi

Notation

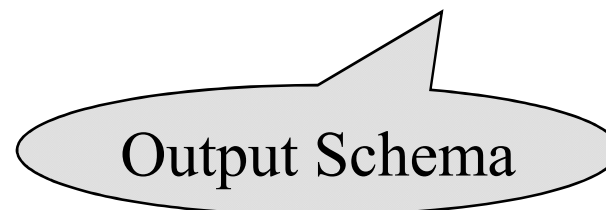


Product(PName, Price, Category, Manufacturer)



```
SELECT pname, price, manufacturer  
FROM   Product
```

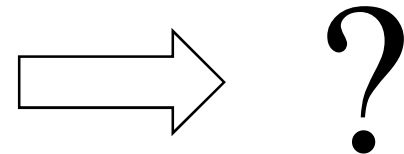
Answer(PName, Price, Manufacturer)



Duplicates from multiple column

Maker	Model	Type
A	1001	pc
A	1002	pc
A	1003	pc
A	2004	laptop
A	2005	laptop
A	2006	laptop
B	1004	pc
B	1005	pc
B	1006	pc
B	2007	laptop
C	1007	pc
...	...	...
....	...	...
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```
SELECT maker, type  
FROM Product
```



Duplicates from multiple column

Maker	Type
A	pc
A	pc
A	pc
A	laptop
A	laptop
A	laptop
B	pc
B	pc
B	pc
B	laptop
C	pc
...	...
....	...
	CSE 303: Ashikur Rahman

```
SELECT maker, type  
FROM Product
```

Duplicates from multiple column

Maker	Type
A	pc
A	pc
A	pe
A	laptop
A	laptop
A	laptop
B	pc
B	pc
B	pc
B	laptop
C	pc
...	...
....	...

SELECT DISTINCT maker, type
FROM Product

Duplicates from multiple column

Maker	Type
A	pc
A	laptop
B	pc
B	laptop
C	pc
...	...
....	...
...	...
....	...
...	...
....	...
...	...
....	...

Simple Predicates

3	SELECT	<attributes>
1	FROM	<one or more relations>
2	WHERE	<conditions>

Semantics

3 **SELECT** $a_1, \dots, a_k$
1 **FROM** R
2 **WHERE** conditions

```
Answer = {}  
for each x in R do  
    if conditions satisfied then  
        Answer = Answer +  $\{(a_1, \dots, a_k)\}$   
return Answer
```

Semantics

3 **SELECT DISTINCT** $a_1, \dots, a_k$
1 **FROM** R
2 **WHERE** conditions

```
Answer = {}  
for each x in R do  
    if conditions satisfied then  
        Answer = Answer  $\cup \{(a_1, \dots, a_k)\}$   
return Answer
```

Number Range Selection

Show all product's info that will cost more than 100 dollars

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

```
SELECT *  
FROM Product  
WHERE price > 100
```

```
Answer = {}  
for each x in R do  
    if conditions satisfied then  
        Answer = Answer + {(a1,...,ak)}  
return Answer
```

PName	Price	Category	Manufacturer
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

Number Range Operators

- Greater than “>”
- Less than “<”
- Greater than or equal to “>=”
- Less than or equal to “<=”
- Two forms of not-equal-to
“!=” and “< >”

BETWEEN-AND operator

```
SELECT <attributes>  
FROM   <table>  
WHERE  attribute BETWEEN value1 AND value 2
```

BETWEEN-AND
is **inclusive**

Works on numbers, texts and dates

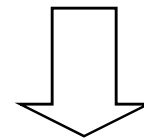
BETWEEN-AND operator

Show all product's info in the price range \$100 to \$200 (inclusive)

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

```
SELECT *  
FROM Product  
WHERE price BETWEEN 100 AND 200
```



BETWEEN-AND
is **inclusive**

PName	Price	Category	Manufacturer
SingleTouch	\$149.99	Photography	Canon

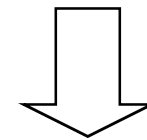
NOT BETWEEN-AND operator

Show all product's info not in the price range \$10 to \$200 (inclusive)

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

```
SELECT *  
FROM Product  
WHERE price NOT BETWEEN 10 AND 200
```



NOT BETWEEN-
AND is also
inclusive

PName	Price	Category	Manufacturer
MultiTouch	\$203.99	Household	Hitachi

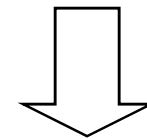
String Matching

Show all household product's info

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

```
SELECT pname, price, manufacturer
FROM Product
WHERE category = 'Household'
```



Matches even category is
fixed length CHAR data type
right padded with blank

PName	Price	Manufacturer
MultiTouch	\$203.99	Hitachi

The **LIKE** operator

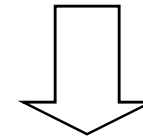
- **s LIKE p**: pattern matching on strings
- **p** may contain two special symbols:
 - (1) **%** → any sequence of characters (zero or more)
 - (2) **_** → any single character (exactly one)

The **LIKE** operator

Show all product's info that has 'mo' as a substring of the product name

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

```
SELECT *  
FROM Products  
WHERE pname LIKE '%mo%'
```



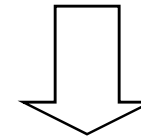
PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks

The **LIKE** operator

Show all product's info where the product name ends with 'Touch'

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

```
SELECT *  
FROM Products  
WHERE pname LIKE '%Touch'
```

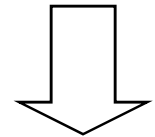


PName	Price	Category	Manufacturer
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

The LIKE operator

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

```
SELECT *  
FROM Products  
WHERE manufacturer LIKE 'G_z_oWorks'
```



PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks

Searching for three special symbols

- ' → 'Baby's World'
- % → '10%'
- _ → 'USA_Canada'

Searching for three special symbols

- ' → 'Baby's World'

Search for all company names having apostrophe s in the name?

Use double apostrophe

```
SELECT *  
FROM Company  
WHERE cname LIKE '%''s%'
```

Searching for three special symbols

- % → '10%'

Search for all interest rates having percent at the end?

Use ESCAPE characters

SELECT *

FROM Bank

WHERE interestRates LIKE '%\%' ESCAPE '\'

Searching for three special symbols

- ‘_’ → ‘USA_Canada’

Search for all country names having underscore inside?

Use ESCAPE characters

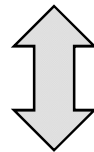
SELECT *

FROM Company

WHERE country LIKE ‘%_%’ ESCAPE ‘\’

LIKE does not pad with blank before matching

```
SELECT pname, price, manufacturer  
FROM Product  
WHERE category = 'Household'
```



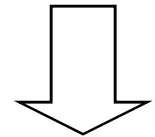
Not equivalent

```
SELECT pname, price, manufacturer  
FROM Product  
WHERE category LIKE 'Household'
```

The **NOT LIKE** operator

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

```
SELECT *  
FROM Products  
WHERE manufacturer NOT LIKE '%mo%'
```



PName	Price	Category	Manufacturer
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

Combining Predicates

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

Show all household products having prices more than \$100

```
SELECT *  
FROM Products  
WHERE category = 'Household'  
AND price >= 100
```

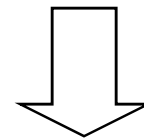
Combining Predicates

Show all manufacturer that make photography or household items

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

```
SELECT DISTINCT maker
FROM   Product
WHERE  category = 'Photography'
OR     category = 'Household'
```



Manufacturer
Canon
Hitachi

Combining Predicates

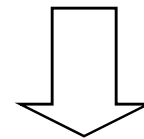
Show all manufacturer that make both photography and household items

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

WRONG!

```
SELECT DISTINCT maker
FROM Product
WHERE category = 'Photography'
AND category = 'Household'
```



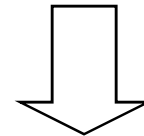
PName	Price	Category	Manufacturer
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Combining Predicates

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

```
SELECT *  
FROM Product  
WHERE pname LIKE '%Giz%'  
OR category LIKE '%hold'  
AND price > 100
```



AND has higher
precedence over **OR**

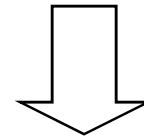
Pname	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
MultiTouch	\$203.99	Household	Hitachi

Combining Predicates

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

```
SELECT *  
FROM Product  
WHERE (pname LIKE '%Giz%'  
OR      category LIKE '%hold')  
AND      price > 100
```



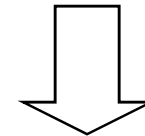
Pname	Price	Category	Manufacturer
MultiTouch	\$203.99	Household	Hitachi

Ordering the Results

Default order is Ascending
Use DESC for descending

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

```
SELECT pName, price, category, manufacturer
FROM Product
WHERE price > 50
ORDER BY pName
```



Pname	Price	Category	Manufacturer
MultiTouch	\$203.99	Household	Hitachi
SingleTouch	\$149.99	Photography	Canon

ORDER BY: Sorting the Results

```
SELECT PName, Price, Manufacturer  
FROM Product  
WHERE category='Gadgets' AND price > 50  
ORDER BY price, pName
```

Ties are broken by the second attribute on the ORDER BY list, etc.

Ordering is ascending, unless you specify the DESC keyword.

ORDER BY: independent ordering possible

```
SELECT PName, Price, Manufacturer  
FROM Product  
WHERE category='Gadgets' AND price > 50  
ORDER BY price ASC, pName DESC
```

NULL values always comes last

You can say

...

ORDER BY price ASC NULLS FIRST

...

ORDER BY price ASC NULLS LAST

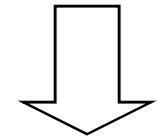
IN operator

CDCollection

Find all albums
launched in 1984 or
1996.

artist	album	tracks	company	year
U2	The Unforgettable	10	Island	1984
U2	Rattle and Hum	17	Island	1988
U2	Achtung Boys	12	Island	1991
Underworld	Second Toughest	8	Junior	1996

```
SELECT album
FROM CDDCollection
WHERE year IN (1984, 1996)
```



```
SELECT album
FROM CDDCollection
WHERE year = 1984
OR year = 1996
```

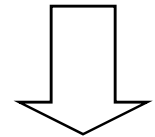
artist	album	tracks	company	year
U2	The Unforgettable	10	Island	1984
Underworld	Second Toughest	8	Junior	1996

NOT IN operator

CDCollection

artist	album	tracks	company	year
U2	The Unforgettable	10	Island	1984
U2	Rattle and Hum	17	Island	1988
U2	Achtung Boys	12	Island	1991
Underworld	Second Toughest	8	Junior	1996

```
SELECT *  
FROM CDCollection  
WHERE year NOT IN (1984, 1996)
```



```
SELECT *  
FROM CDCollection  
WHERE year < > 1984  
AND year < > 1996
```

artist	album	tracks	company	year
U2	Rattle and Hum	17	Island	1988
U2	Achtung Boys	12	Island	1991